Session 2: Shelves, cards and choices

Question: Which order lines are big?

Concepts

Concept	In daily life	In Excel	The limit
list	A row of bins on one shelf	One column of values	The first bin is number 0
len, sum, min, max	Count, total, smallest, largest	COUNT, SUM, MIN, MAX	Same idea, new spelling
dictionary	A product card: field name and value	One row with headers	Look up by name, not position
comparison	"Is this more than that?"	=A2>10	One = stores, two == ask
if, elif, else	A packing rule	=IF(...)	The 4-space indent is part of the rule

Code patterns

```
quantities = [6, 6, 8, 6, 6, 2, 6]
print(quantities[0], quantities[-1])
print(len(quantities), sum(quantities))

order_line = {"quantity": 6, "country": "United Kingdom"}
print(order_line["quantity"])

if quantity >= 100:
    print("Big")
elif quantity >= 12:
    print("Medium")
else:
    print("Small")
```

Three common errors

Error	What to do
IndexError	You asked for a bin that does not exist. The list has 7 items, so the last index is 6.
KeyError	The name is not on the card. Check the spelling of the key.
IndentationError / SyntaxError	A missing colon after if, or the lines under it are not indented.

Exercises and expected answers

Exercise	Expected
quantities[2] and quantities[-1]	8 and 6
Price and country from the card	2.55 and United Kingdom
Label quantity 12, then 100	Medium, then Big
UK or Outside UK rule	UK, then Outside UK for Germany